Riansares 4R — energy efficiency, insulation and ventilation

Which energy improvements should be considered when renovating a home?

Depending on the diagnosis, work on the roof, external walls, openings, windows and doors, heating and cooling systems and ventilation may be appropriate. First review air leakage, damp and the building envelope’s condition. Measures must be designed together and sized for applicable conditions and regulations. IDAE, practical renovation guide.

Can replacing windows increase condensation?

If improved airtightness reduces air renewal and ventilation is not reviewed, indoor moisture balance may change. This does not mean new windows are inherently defective: thermal bridges, temperature and ventilation must be analysed. CTE DB HS regulates aspects of health and hygiene where applicable.

Can a wall with water ingress be insulated?

First identify and control the water source. Installing insulation over persistently damp substrates without a design can conceal damage, hinder drying or create new condensation. Review materials, vapour permeability and use conditions with a competent technical professional.

Does a bathroom renovation need to consider ventilation?

Yes. Moisture generated by a shower or bath requires assessment of air renewal and extraction according to the room’s characteristics. An extractor must be installed and sized according to regulations and actual conditions; it does not replace repairing a leak.

What is the difference between thermal and acoustic insulation?

Thermal insulation reduces heat transfer; acoustic insulation aims to control sound transmission. Systems and performance are not automatically interchangeable, although some solutions provide both. Selection depends on the objective, substrate and applicable requirements.

Hypothetical case E-01 — Very cold bedroom with replacement windows

Fictional example. The home retains uninsulated walls and cold spots appear. Assess air leakage, thermal bridges, ventilation, opportunities to improve the envelope and compatibility with existing damp before recommending a product.